Call for Papers – IEEE SIU 2027

35th IEEE Conference on Signal Processing and Communications Applications
July 4–7, 2027
Istanbul Medipol University, Kavacik South Campus, Beykoz, Istanbul
siu2027.medipol.edu.tr

Important Dates

- **Special Session and Tutorial Proposals Deadline:** November 16, 2026
- **Paper Submission Deadline:** February 1, 2027
- **Notification of Acceptance:** April 30, 2027
- **Camera-Ready Paper Submission and Author Registration Deadline:** May 24, 2027
- **Conference Dates:** July 4–7, 2027

Tracks and Topics of Interest

The conference invites original papers in the following six tracks and in all theoretical and applied areas of signal processing and communications.

Track 1: Communications and Networks

Track Chairs: Berna Özbek (Izmir Institute of Technology), Elif Tuğçe Ceran Arslan (Middle East Technical University), Gökhan M. Güvensen (Middle East Technical University), Özlem Tuğfe Demir (Bilkent University)

- Information Theory and Coding
- Communication Theory and Applications
- Wireless Communications and Networks
- 6G and Beyond Technologies
- Internet of Things (IoT)
- Vehicular Communications
- Cyber-Physical Systems
- Integrated Sensing and Communications
- Satellite and Deep Space Communications
- Optical Communications and Networking
- Edge Computing, Edge Intelligence and Fog Networks
- Energy Harvesting and Low Energy Communications
- Security and Privacy in Communications and Networking
- Digital Twins
- Molecular and Nano Communications
- Quantum Communications
- Age and Value of Information
- Cooperative Communications and Networking
- Energy-Efficient and Green Networking
- Holographic Surfaces and MIMO
- Millimeter-Wave and Terahertz Communications
- Network Security and Privacy
- Physical Layer Security
- Resource Allocation
- Software Defined Networking/Network Functions Virtualization
- Semantic and Goal-Oriented Communication
- UAVs and Non-Terrestrial Networks
- Wireless Power and Information Transfer
- RIS-based Communications

Track 2: Image Processing and Computer Vision

Track Chairs: Hasan F. Ateş (Özyeğin University), Ramazan Gökberk Cinbiş (Middle East Technical University), Selim Aksoy (Bilkent University), Yakup Genç (Gebze Technical University)

- Image and Video Processing
- Object Detection and Pattern Recognition
- Multi-Channel and Multi-Camera Processing
- Image/Video Coding and Compression
- Image and Video-based Biometrics
- Document Analysis and Understanding
- Remote Sensing and Geospatial Analysis
- 3D Imaging and Computational Photography

Track 3: Signal Processing and Robotics

Track Chairs: Elif Vural (Middle East Technical University), Ertan Kazıklı (TOBB University of Economics and Technology), İsmail Uyanık (Hacettepe University), Mert Kalfa (Sabancı University), Ömer Nezih Gerek (Eskişehir Technical University)

- Signal Processing Theory
- Statistical Signal Processing
- Radar Signal Processing
- Signal Processing for Cybersecurity Applications
- Robotics and Automation
- Human-Computer Interaction and Behavior Analysis
- Industrial and Automotive Applications
- Real-Time Signal Processing and Embedded Systems
- E-Health Applications and Assistive Technologies
- Financial Signal Processing
- Marine, Air, and Land Robots
- Industrial Robots
- Humanoid and Social Robots
- Swarm Robots
- Signal Processing for Autonomous Systems

Track 4: Machine Learning and Artificial Intelligence

Track Chairs: Ayşegül Dündar (Bilkent University), Cem Tekin (Bilkent University), Çiğdem Gündüz Demir (Koç University), Hamdi Dibeklioğlu (Bilkent University), Onur Varol (Sabancı University), Tunca Doğan (Hacettepe University), Zafer Doğan (Koç University)

- Machine Learning Theory
- Deep Learning
- Machine Learning for Healthcare
- ML/DL for Communication Systems
- Explainable AI and Trustworthy ML
- Multimodal Analysis
- Reinforcement Learning
- Machine Learning and Deep Learning for Signal and Image Processing Applications
- Adversarial ML and Robust AI
- Transfer, Semi-Supervised and Unsupervised Learning
- Performance Analysis of ML Techniques
- Unsupervised and Generative Models

Track 5: Biomedical Signal/Image Processing

Track Chairs: Aydın Akan (Izmir University of Economics), İmam Şamil Yetik (TOBB University of Economics and Technology), Mustafa Alper Selver (Dokuz Eylül University), Tolga Çukur (Bilkent University), Yeşim Serinağaoğlu Doğrusöz (Middle East Technical University)

- Biomedical Signal Analysis
- Medical Image Analysis and Applications
- Bioinformatics and Genomic Signal Processing
- Applications of AI for Medical Diagnosis
- Biomedical Data Privacy and Security
- Biosignal Processing for Health Monitoring
- Biometric Signal Processing
- Telemedicine and Remote Patient Monitoring
- Neuroengineering and Brain Signal Processing
- Wearable Sensors and E-Health

Track 6: Natural Language Processing

Track Chairs: Çağrı Toraman (Middle East Technical University), Mehmet Fatih Amasyalı (Yıldız Technical University), Murat Can Ganiz (Marmara University)

- Large Language Models and Training Strategies
- Artificial Intelligence and Large Language Model Agents
- Dialogue and Interactive Systems
- Information Extraction and Retrieval
- Linguistic Theories and Cognitive Models
- Translation and Multi-Language Processing

Submission Guidelines

- **Paper Format:** The maximum length of a paper is four pages.
- **Language:** Papers must be written in Turkish. If one of the authors is a non-native speaker of Turkish, papers written in English are considered for review.
- **Review Process:** All submissions will be evaluated with a double-blind review policy. Authors must ensure that their papers are prepared anonymously.
- **Publication:** Accepted and presented papers are planned to be submitted to the IEEE Xplore Digital Library.

For detailed submission instructions, templates, and guidelines, please refer to the official conference website: siu2027.medipol.edu.tr.

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